

INFO

Location

Toronto, ON
Open to relocation
(416) 827-3687

Links

- james.w.spears@gmail.com
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INTRO

About Me

Solution Architect and technical leader with 7+ years designing enterprise application architecture, leading 3rd-party system integrations, and modernizing legacy technology stacks in financial services. Trusted technical advisor to business and technology stakeholders, translating enterprise-level requirements into scalable, AI-enabled solutions and bridging engineering, product, and security teams. Currently the Solution Architect for BMO's Online Banking for Business (OLBB) platform — architecting AI-driven contact centre integrations (Amazon Connect, AWS Bedrock), leading ServiceNow-based customer service workflow integrations, and directing platform-wide modernization across a suite of products serving millions of commercial banking customers. Open to relocation.

EDUCATION

University of Waterloo, Waterloo, ON
Bachelor of Science, General Science,
commencement Fall 2018

Durham College, Oshawa, ON
Advanced Diploma, Electromechanical Engineering Technologist,
commencement Fall 2011

EXPERIENCE

BMO Financial Group

BMO Financial Group is the 8th largest bank in North America, serving over 12 million customers across a highly diversified range of financial services.

Solution Architect - [BMO Financial Group](#) Jan 2022 - Present
([OLBB](#)), Toronto, ON

- Serve as trusted technical advisor to business and technology stakeholders, defining architecture patterns and translating enterprise requirements into scalable platform solutions across the OLBB commercial banking platform
- Own end-to-end architecture and delivery of Angular front-end applications and their Node.js backend APIs, including data integrations across multiple systems and 3rd-party platforms
- Directed the OLBB-wide Angular upgrade programme — the largest front-end modernisation effort across the platform, migrating 30+ applications spanning AngularJS, Angular 4, and Angular 7 all the way to Angular 19;

authored the foundational libraries and SDKs that made incremental migration possible, then planned and led the full rollout across every UI surface

- Tech lead on the OLBB *Getting Started Hub* — a content/document delivery platform designed to accelerate new commercial banking customer onboarding
- Led engineering delivery of *Lexicon* — a dual-format Angular and Web Component design system and reusable component library for commercial banking, which earned the **Red Dot Award (2024)** and is now the standard UI layer across the OLBB product suite
- Architected AI-driven contact-centre integrations (Amazon Connect, AWS Bedrock GenAI) for commercial banking customer support, including a TransUnion caller-verification service for fraud detection and a Contact Lens transcript-analysis pipeline for automated call quality and insights — delivering measurable business outcomes while integrating cleanly with existing systems
- Led technical discovery and integration of OLBB applications with ServiceNow, a 3rd-party enterprise workflow and document/case management system, including delivery of a virtual-agent chatbot for commercial banking customer support
- Recruit, mentor, and grow a team of full-stack engineers; established code-review practices, architectural standards, and a culture of continuous improvement

WebsiteBox

WebsiteBox is a SaaS platform providing website hosting and WYSIWYG design tools to real estate agents and brokerages.

Node.js Developer - [WebsiteBox](#), Toronto, ON Feb 2021 - Dec 2021

- Built and maintained Angular and AngularJS front-end applications backed by Node.js/Express microservices
- Drove the migration from a legacy billing system to Stripe, delivering reliable, PCI-compliant payment processing
- Worked across a polyglot data layer including Elasticsearch, Apache Cassandra, and MySQL (MariaDB)

FliteX

FliteX is an early-stage startup building a SaaS platform for AI-driven airline route optimization.

Full Stack Developer - [FliteX](#), Oshawa, ON March 2020 - June 2020

- Prototyped a geospatial SPA (Angular) for interactive flight route visualization and optimization
- Ported core algorithmic models from MATLAB to Python (NumPy, SciPy), improving iteration speed and testability

Wysdom AI

Wysdom AI is an AI company offering enterprise virtual assistant and chatbot software via a SaaS platform, primarily serving large telecoms.

AI Automation Specialist - [Wysdom AI](#), Richmond Hill, ON June 2019 - March 2020

- Designed and implemented automated chatbot regression testing in Python, reducing manual QA effort

- Built web-based business process automation tools that improved internal operational workflows
- Contributed to a prototype audio-processing data pipeline using FFmpeg for ML training data preparation

SKILLS

- | | | |
|--------------------------------|------------------------------------|----------------------------|
| • RESTful API Design | • AWS Cloud Architecture | • Agile / Scrum |
| • Microservices & Integrations | • AWS Bedrock / GenAI | • Angular / Web Components |
| • Node.js / Express | • Contact Centre & CX Architecture | • Kubernetes / OpenShift |
| • TypeScript / JavaScript | • Amazon Connect / Contact Lens | • React |
| • ServiceNow Integrations | • CI/CD and IaC | • HTML / CSS |
| | | • Python |

CERTIFICATIONS

- AWS Certified Solutions Architect Professional, *SAP-C02*, [Verify](#)
- OpenJS Node.js Services Developer, *JSNSD*, [Verify](#)
- OpenJS Node.js Application Developer, *JSNAD*, [Verify](#)
- AWS Certified Solutions Architect (Associate), *SAA-C02*, [Verify](#)
- AWS Certified Cloud Practitioner, *CLF-C01*, [Verify](#)
- Certified Associate in Python Programming, *PCAP-31-02*, [Verify](#)

PROJECTS

- Secrets-as-Code CLI - ([Demo](#)) — Secrets-as-code CLI tool; built end-to-end using TDD and BDD with AI-assisted development (Claude Code)
- Video Summarizer - ([Demo](#)) - Alpine.js, TailwindCSS for styling, WASM (if available), and OpenAI API

EXTRA-CURRICULARS

- Waterloo - Team Leader Magnetic Wheel. Competed at SpaceX [Hyperloop Pod Competition](#) 2017.

PUBLICATIONS

- [Approaching special relativity in complex time](#), Can. J. Phys., **95(10)** (2017) 923-926., Mathematical/theoretical physics, logic, set theory, and algebra

AWARDS

- [Pascal Mathematics Contest](#), School champion, 25th percentile.